ARDHI UNIVERSITY

STUDENTS COURSE EVALUATION MANAGEMENT INFORMATION SYSTEM (CEMIS)

User Manual

May, 2023

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INTRODUCTION

1.1 Background and Context

Like any other university, Ardhi University has three core functions: teaching, research, and consulting. To ensure that the University carries out its operations in a manner acceptable to the Senate and the accrediting authority, such as the Tanzania Commission for Universities (TCU), it has set out some organs in its administrative structure to oversee the quality of delivery of the core functions. One of the organs is the Quality Assurance Unit (QAU). One of the responsibilities of the QAU is to ensure that ARU's curricula meet TCU's standards and that both staff and students follow the teaching schedule to ensure effective learning. The QAU has therefore developed a procedure that enables students to assess staff performance and the relevance of their enrolled course by evaluating them at the end of each semester.

Currently, the University conducts staff evaluations and suggestions to improve the programmes manually. The QA unit provides forms for students to complete and process manually. Such practice has several weaknesses, including students not returning the forms or returning them late. Moreover, the QA unit spends a lot of time processing the manually filled forms, as one has to first enter all the documents in Excel or any other statistical software before processing them, causing difficulties in preparing desired reports.

To enhance the student course evaluation process, the University, through the HEET project, has decided to automate the entire process by developing a web-based system for handling the evaluation process. The system provides options for data entry, processing, and preparing different reports related to quality assurance issues.

1.2 Objective of CEMIS System

The primary purpose of having the Student Course Evaluation Management Information System (CEMIS) is to automate the student evaluation process. The automation process uses a web-based system that allows staff and students to perform the evaluation more robustly. Specifically, the online system can:

i) Give a user-friendly interface for data entry, processing, and generation of various reports as per Quality Assurance Unit requirements;

ii) Provide users access to student and staff data from AMIS to create reports that can inform decision-making on teaching methods, styles, and course content relevance based on student feedback.

With the CEMIS system, students can evaluate their lecturers' teaching methods and styles and provide feedback on curricula. Such feedbacks include comments and suggestions on various aspects such as course content, teaching methods, reference materials, and how relevant the course is to their future careers.

The University developed the CEMIS system to address challenges faced using the traditional manual evaluation process. The idea was to automate the current practice to improve its efficiency, save time, and ensure the accuracy and consistency of evaluation reports.

1.3 CEMIS Modules

The CEMIS system consists of three modules. The first module is Teaching Methods and Styles, the second is Suggestion and Recommendation, and the third is User Authentication and Authorisation.

The Teaching Methods and Style module is developed to allow students to evaluate lecturers in terms of the way they teach and how they use different media to help students grasp the materials being delivered. Students evaluate their lecturers' use of teaching aids and media such as audio, visual, PowerPoint slides, videos, etc. Students also gauge their lecturers in terms of attendance, organization of the lecturer's materials, and coverage of the course content.

The module on Suggestions and Recommendations allows students to commend the usefulness of course contents, reference materials, practical exercises, etc. Using this module; students can suggest improvements to the course in terms of concepts, teaching methods, reference materials, practical, or seminars, and the relevance of the course to prospective careers. Students are required to evaluate at least three courses per semester to inform the management of the usefulness of the curricula and their satisfaction with the courses they are enrolled in.

In addition, the module for User Authentication and Authorisation was developed to help system administrators authenticate valid users using their usernames and passwords from the AMIS system. This means that the Student Course Evaluation System users do not need to create new usernames and passwords. They will use the same usernames and passwords from AMIS to access and navigate the CEMIS system. Moreover, the User Authorisation part of the module allows the system administrators to set privileges for different system users. For

instance, students will be given access to enter data and carry out assessments but cannot read the reports. They will also not be allowed to change the reviews once they have submitted the data. Staff from Quality Assurance Unit will have read-only privileges. Therefore, they can only read and extract evaluation reports from the system without changing anything. They will export data to PDF, Excel, or even MS Word format for further processing.

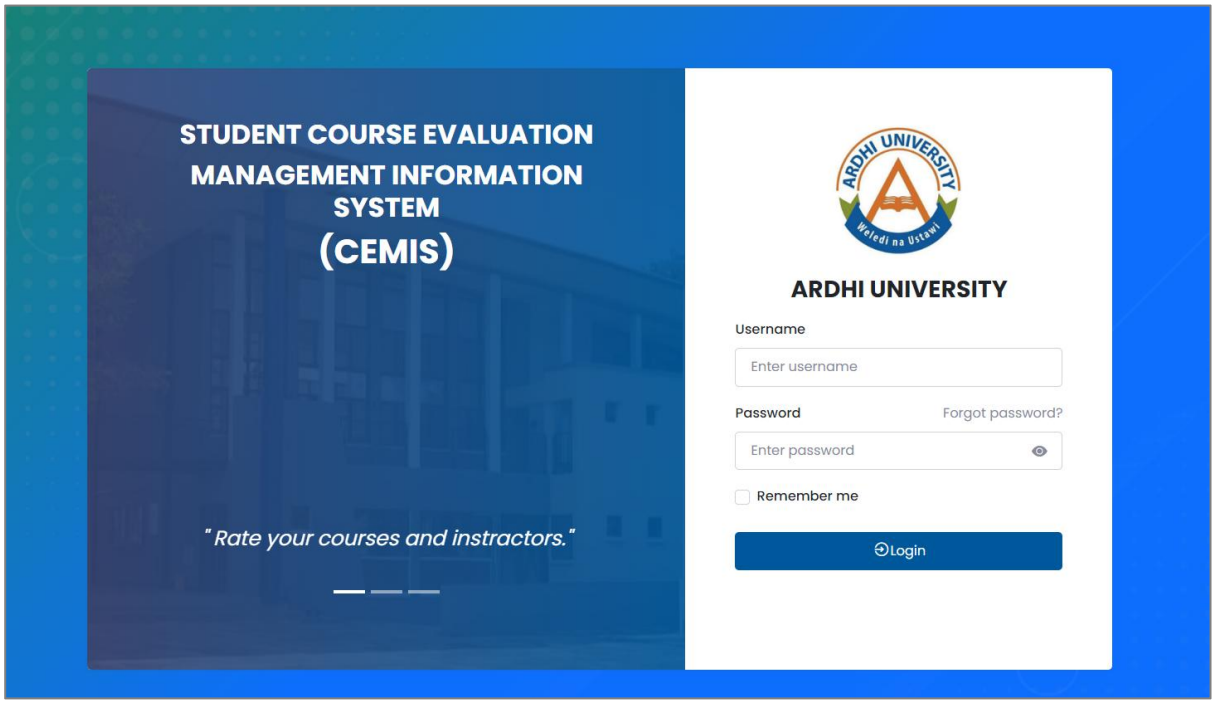
1.4 Need for the CEMIS System User Manual

The purpose of this manual is to assist CEMIS users in comprehending the various functions available and how to access them. This will facilitate entering data, creating reports, and producing results. Moreover, while creating applications, it may be necessary to access certain functions. As this is an online system, a user manual is available in the help function to provide assistance whenever needed. The manual includes detailed, step-by-step instructions to enable users to access various system functions and utilize them independently

CEMIS SYSTEM

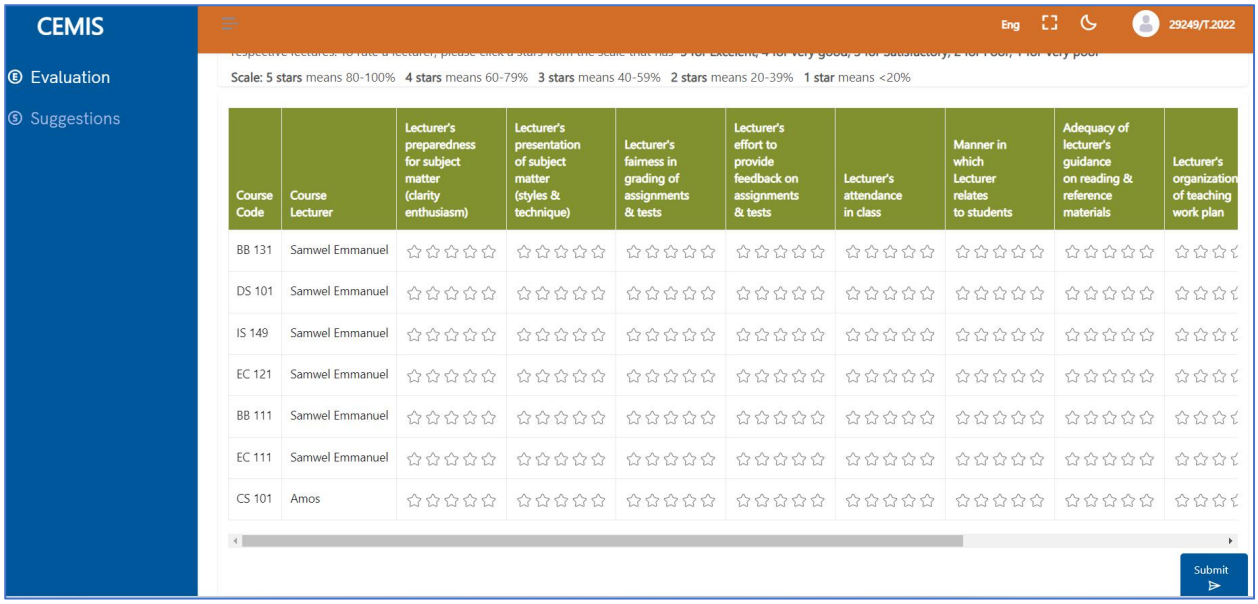
2.1 Access to CEMIS

To access the main CEMIS interface, users must visit the Ardhi University website at www.aru.ac.tz. While on the ARU website, users can click the word CEMIS on the main menu or use the link provided under the related links. Upon clicking on the link, a window like this will appear.



2.2 To perform Staff Evaluation – Teaching Methods and Styles

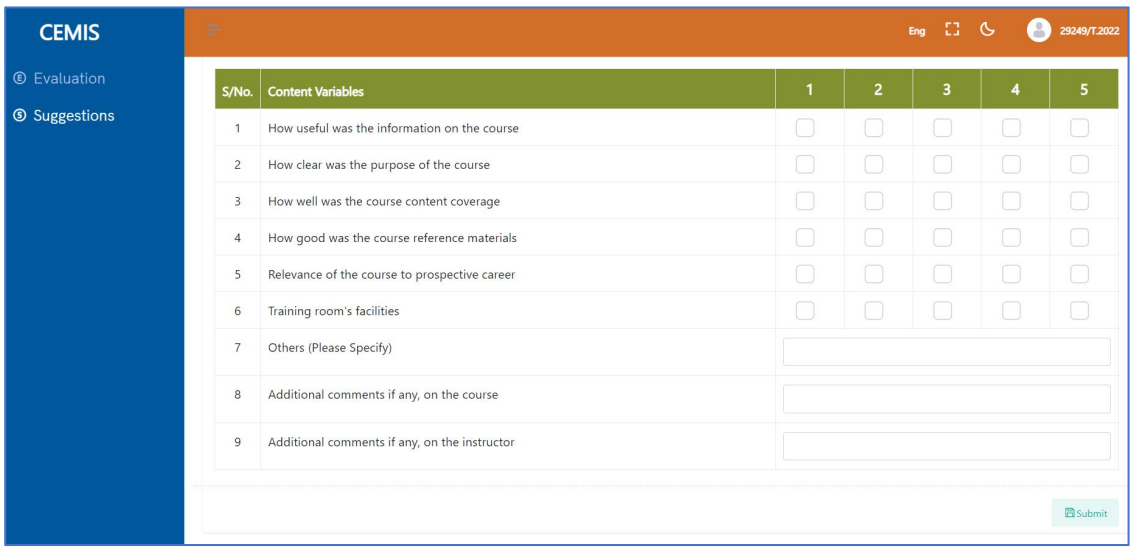
The evaluation system consists of two parts, each with a separate interface. One part is for students who conduct evaluations, and the other is for staff who generate evaluation reports. To access the evaluation page, students must log in to the system using the same username and password they use for AMIS. After logging in, students will see a window that prompts them to assess all staff members across all courses before submitting the assessment form.



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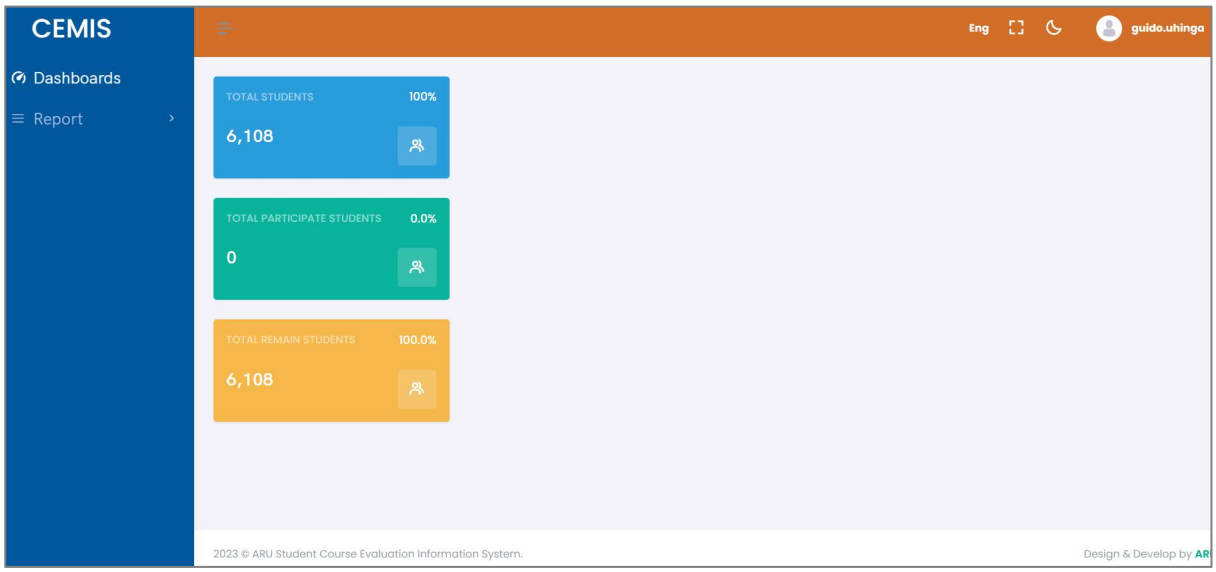
2.3 Suggestions and Recommendations

Students can offer suggestions and recommendations regarding course improvement using the provided rating scales. They must provide comments and suggestions for at least three courses by selecting the suggestion option. Clicking on the suggestion option will open a window for submitting comments.



2.4 Interface for Staff

The system also provides an interface for staff from the QA unit to view and process evaluations done by the students. When staff members enter the CEMIS system using their credentials, the following dashboard will appear:



In this dashboard window, you will view the total number of students registered in AMIS and those who have already completed evaluations for the current semester. Moreover, you will see the number of students still needing to conduct their evaluations. However, users can assess two main types of reports: Those related to teaching methods and styles and those concerned with comments and suggestions that students give to courses.

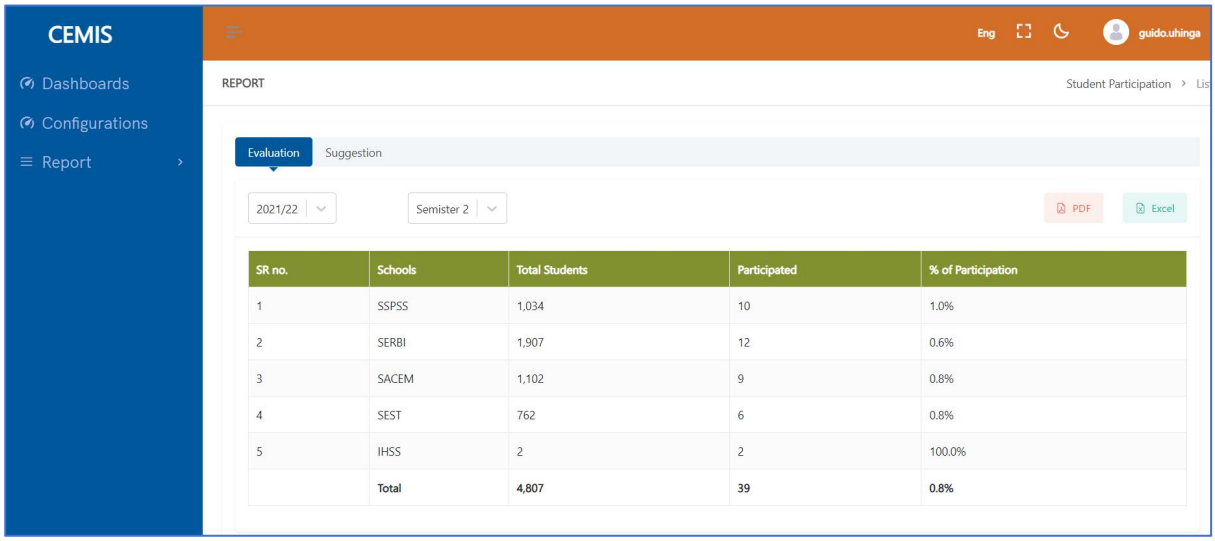
2.5 Evaluation Reports

As mentioned above, the CEMIS system gives students five evaluation reports to assess their lecturer's performance. To access such reports, users must click Reports, followed by the specific type of report they want to get information about. The reports can be accessed using a dashboard by clicking a desired report name. The dashboard provides a Report Module with evaluation reports, suggestions, and recommendations. The five reports that evaluate the performance of lecturers based on their teaching methods and styles are: i) level of participation, ii) overall performance, iii) average rating, iv) average performance, and v) average rating for each lecturer. To access such reports, users must click on Reports, then point to one of the report templates and select Evaluation.

2.5.1 Report on Level of Participation

This report presents the number of students in each school per semester and those participating in the evaluation process. Therefore, to access such a report user must select Evaluation followed by a study

year and semester. For instance, after choosing such fields, under a report on the level of participation, the window presented in Figure 1 appears.

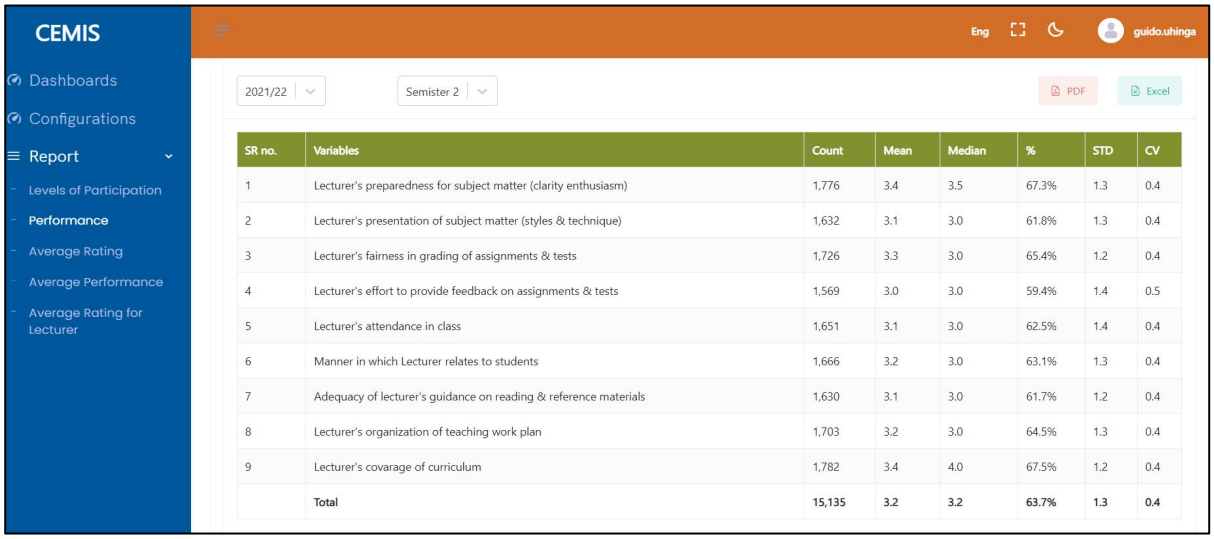


SR no.	Schools	Total Students	Participated	% of Participation
1	SSPSS	1,034	10	1.0%
2	SERBI	1,907	12	0.6%
3	SACEM	1,102	9	0.8%
4	SEST	762	6	0.8%
5	IHSS	2	2	100.0%
	Total	**4,807**	**39**	**0.8%**

Figure 1. Report on Level of Participation

2.5.2 Report on Level of Performance – University-wide

This report shows the general University's evaluation status across nine areas, as presented in Figure 2. The results are presented in terms of count, mean, median, percentage, standard deviation and coefficient of variation. To view this report, one must select a year and semester.

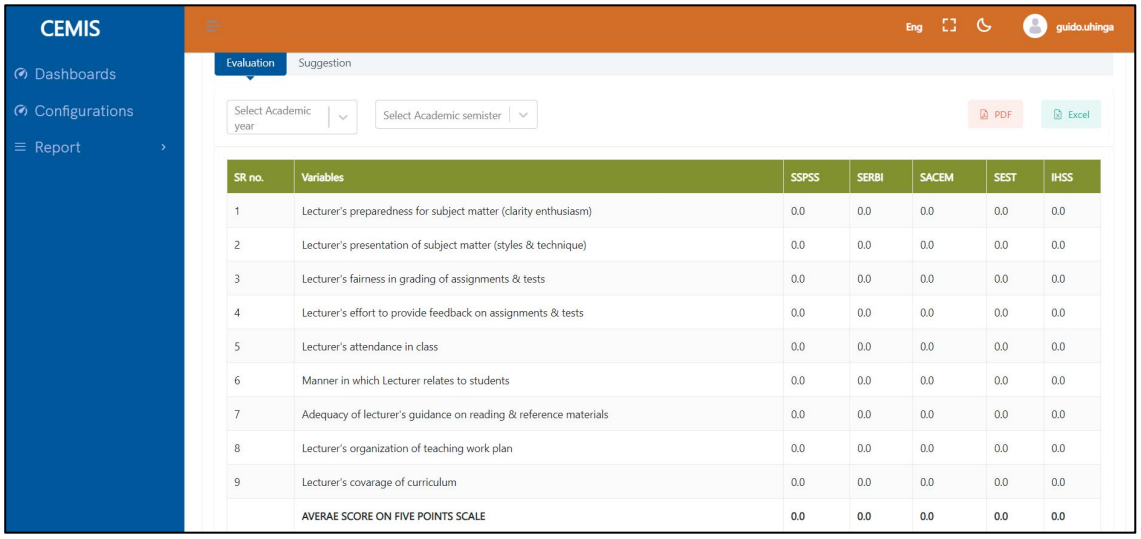


SR no.	Variables	Count	Mean	Median	%	STD	CV
1	Lecturer's preparedness for subject matter (clarity enthusiasm)	1,776	3.4	3.5	67.3%	1.3	0.4
2	Lecturer's presentation of subject matter (styles & technique)	1,632	3.1	3.0	61.8%	1.3	0.4
3	Lecturer's fairness in grading of assignments & tests	1,726	3.3	3.0	65.4%	1.2	0.4
4	Lecturer's effort to provide feedback on assignments & tests	1,569	3.0	3.0	59.4%	1.4	0.5
5	Lecturer's attendance in class	1,651	3.1	3.0	62.5%	1.4	0.4
6	Manner in which Lecturer relates to students	1,666	3.2	3.0	63.1%	1.3	0.4
7	Adequacy of lecturer's guidance on reading & reference materials	1,630	3.1	3.0	61.7%	1.2	0.4
8	Lecturer's organization of teaching work plan	1,703	3.2	3.0	64.5%	1.3	0.4
9	Lecturer's covarage of curriculum	1,782	3.4	4.0	67.5%	1.2	0.4
	Total	**15,135**	**3.2**	**3.2**	**63.7%**	**1.3**	**0.4**

Figure 2. Report on Performance – University-wide

2.5.3 Report on Average Rating – University-wide

This report presents the average rating of the schools based on the nine variables as presented in Figure 3. The report is similar to the one shown in Figure 2, only that this summarises information at a school level. To view the details for each school, follow the same procedure and steps as in Figure 2.

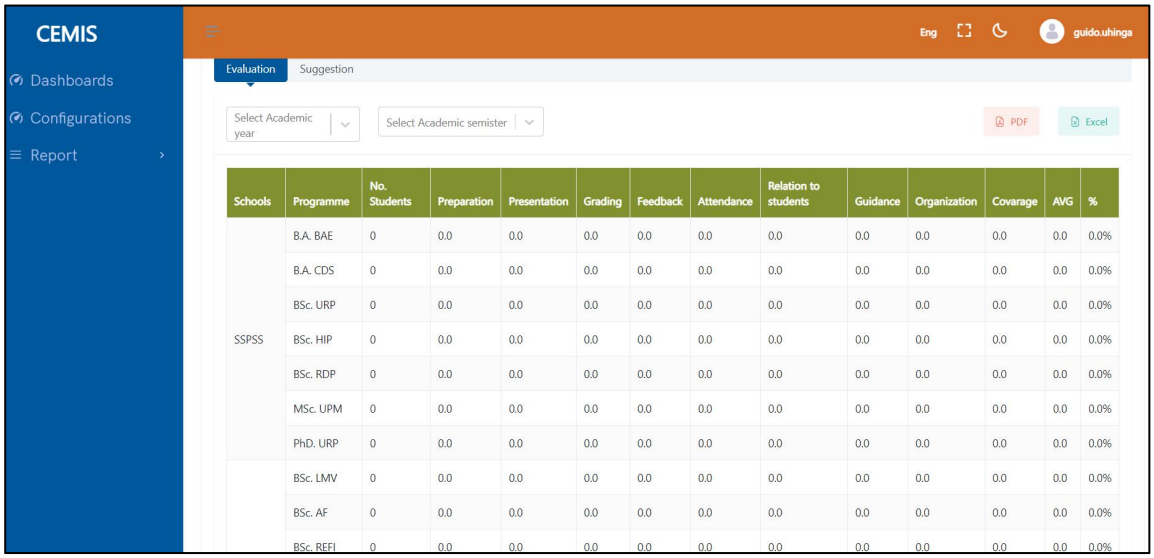


SR no.	Variables	SSPSS	SERBI	SACEM	SEST	IHSS
1	Lecturer's preparedness for subject matter (clarity enthusiasm)	0.0	0.0	0.0	0.0	0.0
2	Lecturer's presentation of subject matter (styles & technique)	0.0	0.0	0.0	0.0	0.0
3	Lecturer's fairness in grading of assignments & tests	0.0	0.0	0.0	0.0	0.0
4	Lecturer's effort to provide feedback on assignments & tests	0.0	0.0	0.0	0.0	0.0
5	Lecturer's attendance in class	0.0	0.0	0.0	0.0	0.0
6	Manner in which Lecturer relates to students	0.0	0.0	0.0	0.0	0.0
7	Adequacy of lecturer's guidance on reading & reference materials	0.0	0.0	0.0	0.0	0.0
8	Lecturer's organization of teaching work plan	0.0	0.0	0.0	0.0	0.0
9	Lecturer's covarage of curriculum	0.0	0.0	0.0	0.0	0.0
	AVERAE SCORE ON FIVE POINTS SCALE	0.0	0.0	0.0	0.0	0.0

Figure 3. Report on Performance – School-wide

2.5.4 Report on Average Performance

The performance report gives an evaluation of the degree programmes offered by each school. It uses twelve variables, which are as in Figure 4. To see specific details for a particular school, follow the steps outlined in Figure 2.



Schools	Programme	No. Students	Preparation	Presentation	Grading	Feedback	Attendance	Relation to students	Guidance	Organization	Covarage	AVG	%
	B.A. BAE	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0%
	B.A. CDS	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0%
	BSc. URP	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0%
SSPSS	BSc. HIP	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0%
	BSc. RDP	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0%
	MSc. UPM	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0%
	PhD. URP	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0%
	BSc. LMV	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0%
	BSc. AF	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0%
	BSc. REFI	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0%

Figure 4. Report on Average Performance

2.5.5 Report on Average Rating for Lecturer

This report presents information about the performance of individual lecturers University-wide. For instance, an average performance report for a particular lecturer (Figure 5) will appear after filling out the appropriate fields such as year, semester, and degree programme. For instance, Figure 5 presents a sample report for the academic year 2022/23 semester 2 for BSc. AF.



Lecturer's Name	Course Code	Respondents	Preparation	Presentation	Grading	Feedback	Attendance	Relation to students	Guidance	Organization	Covarage	AVG	%
Ardolf Bahati Tweve	AC 281	1	4.0	1.0	5.0	4.0	3.0	2.0	4.0	4.0	4.0	3.4	68.9%
Ardolf Bahati Tweve	AC 291	1	5.0	1.0	5.0	3.0	4.0	4.0	3.0	3.0	5.0	3.7	73.3%
N/A	AC 372	1	3.0	5.0	3.0	3.0	3.0	5.0	3.0	5.0	4.0	3.8	75.6%
N/A	AC 377	1	5.0	2.0	4.0	2.0	4.0	4.0	4.0	4.0	5.0	3.8	75.6%
Ardolf Bahati Tweve	BB 221	1	5.0	1.0	1.0	2.0	5.0	4.0	2.0	4.0	4.0	3.1	62.2%
Ardolf Bahati Tweve	BB 232	1	2.0	5.0	2.0	5.0	2.0	3.0	5.0	4.0	2.0	3.3	66.7%
Ardolf Bahati Tweve	BB 233	1	3.0	3.0	3.0	3.0	3.0	2.0	2.0	5.0	3.0	3.0	60.0%
Ardolf Bahati Tweve	BB 241	1	1.0	3.0	4.0	4.0	4.0	2.0	5.0	3.0	2.0	3.1	62.2%

Figure 5. Report on Average Rating for Lecturer

2.6 Reports on Suggestions and Recommendations

The dashboard provides information on teaching methods and styles and allows viewing of student feedback and suggestions regarding their enrolled courses. The reports on recommendations for the courses offered come in four different types. Such reports are i) level of participation, ii) performance, iii) average rating, and iv) average performance.

2.6.1 Level of Participation

Table 6 presents a report showing the number of students participating in the evaluation process. It is similar to the level of participation given in evaluating academic staff members. Users must select Reports, Level of Participation, and Suggestions to produce such reports.

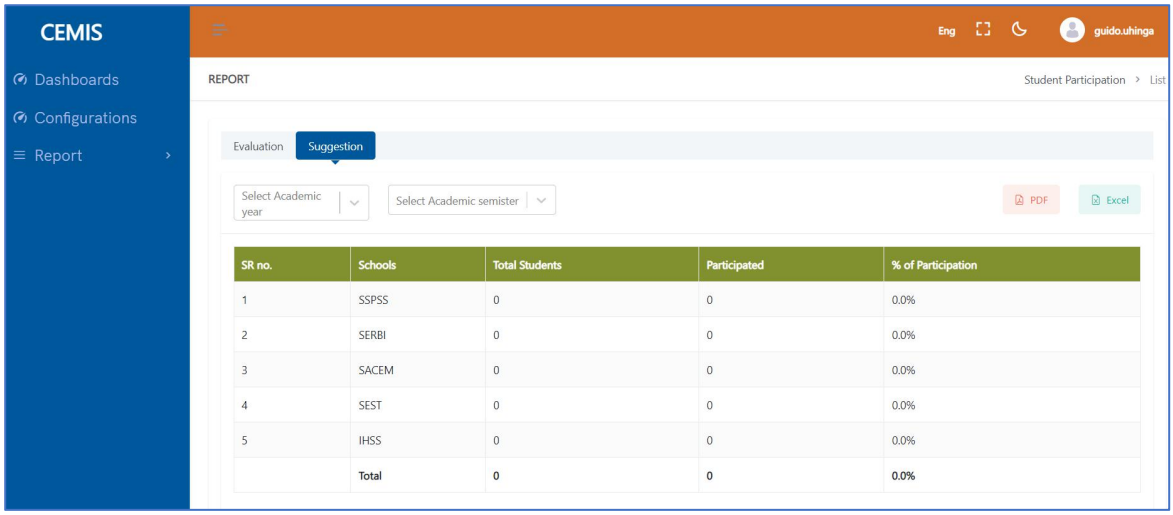


Figure 6. Report on Level of Participation on Suggestions and Recommendations

2.6.2 Performance

The performance presents information on students' recommendations regarding six variables measured using count, mean, median, percentage, standard deviation and coefficient of variation. One must select Reports, Performance, and Suggestions to view such information. A sample of such reports is as in Figure 7.

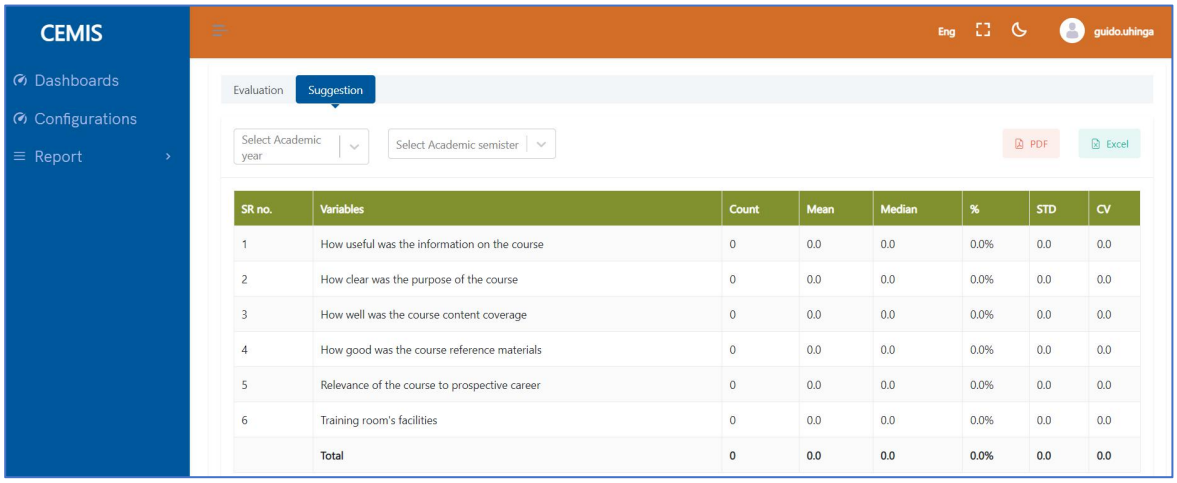


Figure 7. Courses Performance as Recommended by Students

2.6.3 Average Rating

Figure 8 shows the rating of the courses as recommended by students. Users must click Reports, Rating, and then Suggestions to view the rating reports.

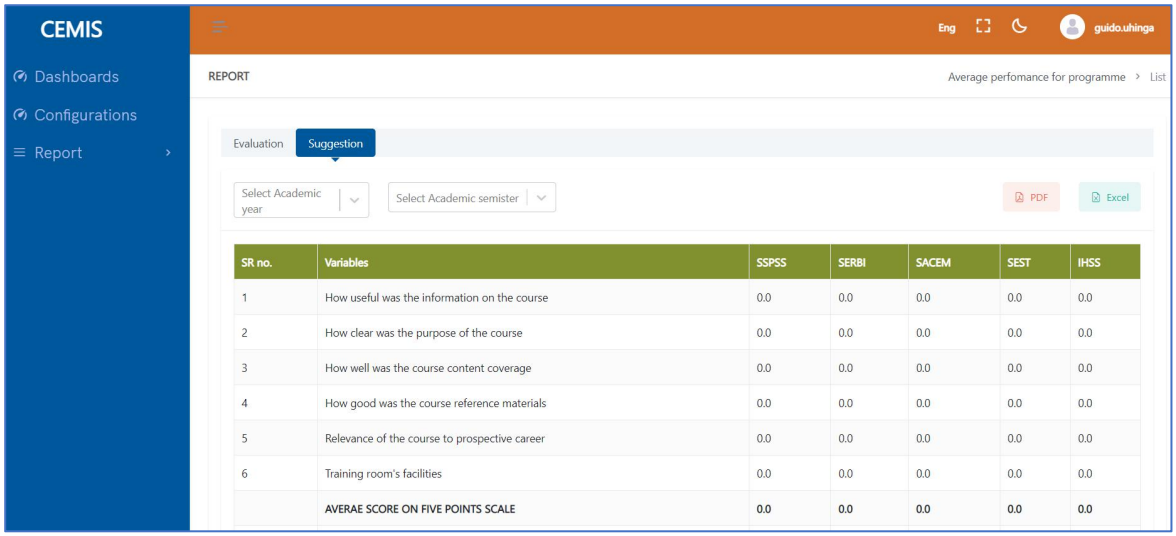


Figure 8: Courses Rating by Students

2.6.4 Average Performance

The average performance reports present information on students' course assessments regarding their relevance and purpose, course information, course content coverage, reference materials and facilities available n the training rooms. To view such reports, one must select Reports, Average Rating, and then Suggestions (Figure 8).

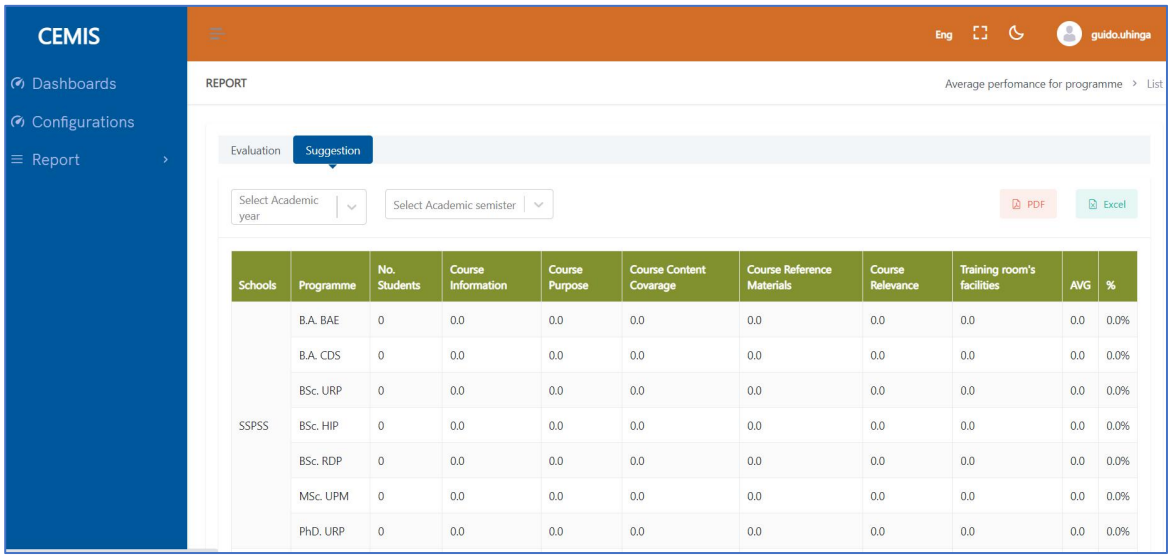


Figure 8: Average Course Performance as Evaluated by Students

2.7 Authentication and Approval

The dashboard provides a configuration module that authenticates and approves CEMIS users. Therefore, while in CEMIS, users send a request to AMIS, and once checked and found to be genuine users, information is sent back to CEMIS for approval. CEMIS users can only access the system after

being registered with AMIS. Therefore, all academic staff members must be registered with AMIS and assigned courses. Moreover, all students must have valid AMIS accounts for them to be able to access CEMIS and perform the assessment. Other evaluation reports can also be accessed using the same approach.